

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VIII
PAPER –2
TEST DATE: 14-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

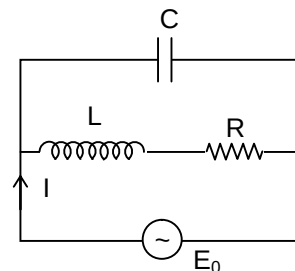
1. For the circuit shown, angular frequency ' ω ' of source is adjusted so that current I through source is in phase with applied voltage of source, then choose the correct alternative(s).

(A) $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$

(B) $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

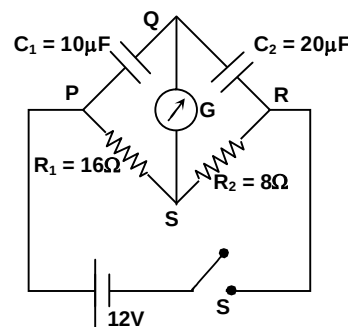
(C) $I = \frac{CE_0 R}{L}$

(D) $I = \frac{2CE_0 R}{L}$



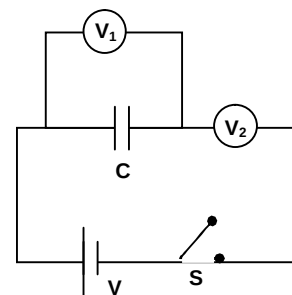
2. In the circuit drawn below if the resistance of galvanometer 'G' is 8Ω , then choose the correct option(s) after long time when switch 'S' is closed.

- (A) Charge on C_1 is $80\ \mu\text{C}$
 (B) Charge on C_2 is $40\ \mu\text{C}$
 (C) Current flowing through R_1 is $0.5\ \text{A}$
 (D) No current flows through the galvanometer 'G'



3. An uncharged capacitor of capacitance ' C ' is connected to two voltmeters V_1 and V_2 . The resistances of voltmeter V_1 and V_2 are $2R$ and R respectively. If switch 'S' is closed at $t = 0$, then

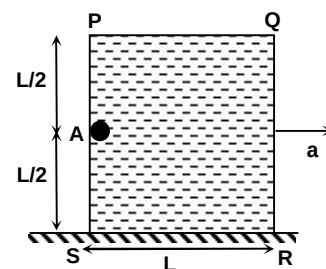
- (A) At $t = 0$, reading of voltmeter V_2 is V
 (B) Reading of V_2 varies from $\frac{4V}{3}$ and $\frac{V}{2}$
 (C) Reading of voltmeters will be equal at $t = \frac{2RC}{3} \ln 2$
 (D) Reading of voltmeter will be equal at $t = \frac{4RC}{3} \ln 2$



4. The radius of the orbit of an electron in hydrogen like atom is $4a_0$ where a_0 is the Bohr radius of hydrogen in ground state. Its orbital angular momentum is $\frac{2h}{\pi}$. It is given that h is plank's constant and R is Rydberg constant. The possible wavelength(s), when the atom de-excites is(are)

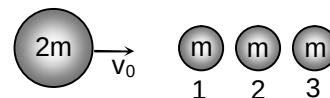
- (A) $\frac{1}{3R}$
 (B) $\frac{1}{6R}$
 (C) $\frac{1}{15R}$
 (D) $\frac{9}{7R}$

5. A small solid ball of density ρ is held inside at point A in a closed cubical container of side L , filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration a horizontally and the ball is released from A simultaneously. It has been observed that the ball hits the point Q after time T . Points A, P, Q, R and S are lying in the same plane. Then choose the correct option(s).



- (A) $a = 2g$
 (B) $a = \frac{3g}{2}$
 (C) $T = \sqrt{\frac{L}{3g}}$
 (D) $T = \sqrt{\frac{2L}{3g}}$

6. A steel ball of mass $2m$ collides elastically with a row of three steel balls each of mass m . If ball of mass $2m$ has initial velocity v_0 and three balls with small separation were at rest. Then after all possible head on collisions.



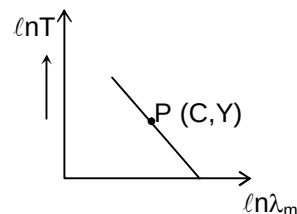
- (A) Ball of mass $2m$ moves towards right with velocity one fourth that of ball 1.
 (B) Ball 1 and ball of mass $2m$ would remain at rest.
 (C) Ball 2 moves with velocity $\frac{4v_0}{9}$ towards right
 (D) Ball 3 moves towards right with velocity $4v_0$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

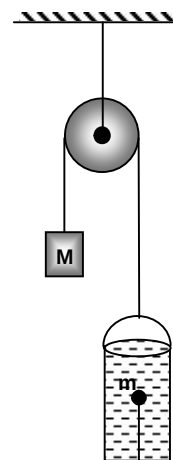
The following graph shows the variation of $\ln T$ versus $\ln \lambda_m$, where λ_m is wavelength corresponding to which maximum number of photons are emitted and T is temperature in Kelvin. x-coordinate of a point on line is C and the intercept made by line 1 on y-axis has length A .



7. What is value of Wien's displacement constant?
 - (A) e^A
 - (B) $\frac{1}{e^A}$
 - (C) $\ln A$
 - (D) $\frac{1}{\ln A}$
8. A black body is emitting radiations at certain temperature. If the radiation emitted by the body is represented by point P on the line, then the value of Y (ordinate of point P) is
 - (A) A
 - (B) C
 - (C) $A-C$
 - (D) $C-A$

Paragraph for Question Nos. 09 and 10

A cylindrical bucket with water in it balances a mass M over a pulley. A piece of cork of mass m and specific gravity σ , is then ties to the bottom of the bucket with the help of spring so as to be touching immersed.



9. Acceleration of the cork is
 (A) g
 (B) $\frac{mg}{m+M}$
 (C) $\frac{mg}{2m+M}$
 (D) $\frac{mg}{m+2M}$
10. Tension in the string connecting cork and cylinder is
 (A) zero
 (B) $mg\left(\frac{1}{\sigma}-1\right)$
 (C) $\frac{2mMg}{(m+2M)}\left(\frac{1}{\sigma}-1\right)$
 (D) $\frac{2mMg}{(2m+M)}\left(\frac{1}{\sigma}-1\right)$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

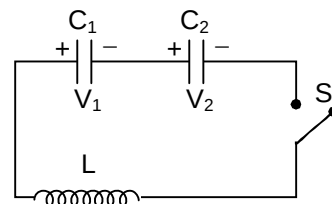
11. A solid body starts rotating about a stationary axis with angular acceleration $\beta = at^2$, a is positive constant and t is time in seconds. After few moments, at time $t = \left(\frac{\alpha \tan \theta}{a}\right)^{1/4}$. It has been found that for a general point on body the velocity vector and acceleration vector form an angle θ with each other. Find the value of α .
12. The principal section of glass prism of isosceles ΔPQR with $PQ = PR$. The face PR is silvered. A ray is incident perpendicularly on face PQ and after two reflections it emerges from base QR , normal to it. If the angle (in degree) of the prism is θ . Find the value of $\frac{\theta}{3}$.
13. An object is placed at distance 20 cm from an equi-concave lens of focal length 30 cm whose one side is silvered. If a virtual image is formed at $\frac{20}{3}$ cm from lens then what is the radius of curvature of the curved surface in cm?

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

Two capacitors shown in the circuit have capacitance $C_1 = 10 \mu\text{F}$ and $C_2 = 30 \mu\text{F}$ and they have charged to potentials $V_1 = 8\text{V}$ and $V_2 = 12\text{V}$ respectively. Switch S is closed to connect them to the inductor $L = 3\text{mH}$.



14. Potential difference across C_2 when the current in inductor is maximum (in volt) is.....
15. The maximum current through inductor in (milli ampere) is

Question Stem for Question Nos. 16 and 17
Question Stem

A toy train in rail museum runs on an elliptical orbit having major and minor axis in the ratio of 4 : 3. The length of the train is exactly equal to half the perimeter of the elliptical track. The train is travelling at a constant speed of 20 m/s. Engine of toy train blows horn on frequency 1730 Hz when its acceleration is minimum. Speed of sound in air is 330 m/s. The frequency of whistle observed by a kid in last compartment of toy train is X Hz and frequency of whistle observed by a kid in central compartment of train is Y Hz.

16. The magnitude of X is
17. The magnitude of Y is

Question Stem for Question Nos. 18 and 19
Question Stem

A thin uniform shell of radius $R = 4\text{m}$ bored such that the axis of the boring rod passes through the centre of spherical shell. The boring rod is a cylinder of radius 2m. Mass of the sphere before boring is $\sqrt{3} \text{ kg}$.

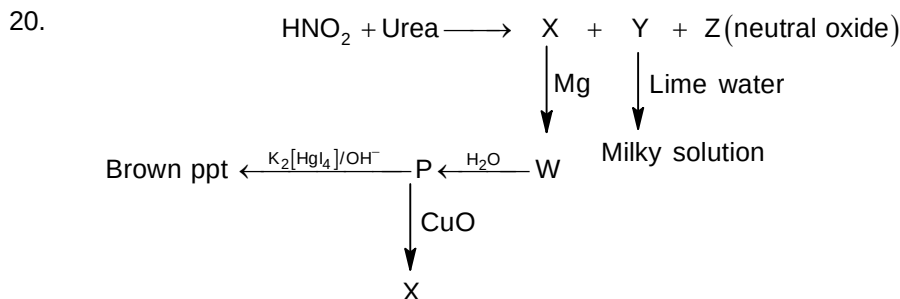
18. The mass of the leftover part (in kg) is
19. The moment of inertia of the leftover part about the axis of the boring rod (in SI unit) is

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).



Which of the following compound have two π – bond in its structure?

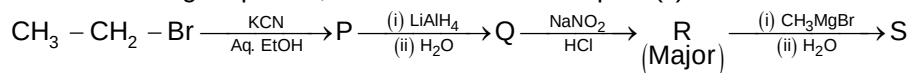
- (A) X
(B) Y
(C) W
(D) P

21. The correct statement (s) about Cr^{2+} and Mn^{3+} are:

(Atomic number of Cr = 24, Mn = 25)

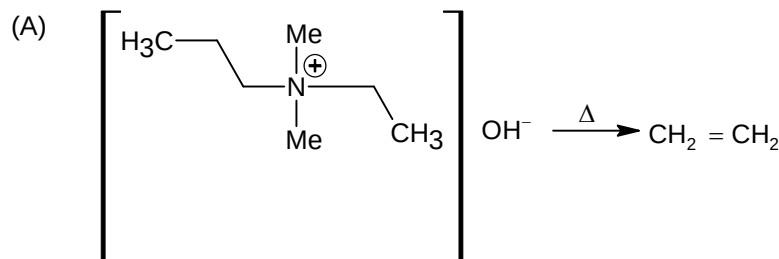
- (A) Cr^{2+} is a reducing agent
(B) Mn^{3+} is an oxidizing agent
(C) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration
(D) When Mn^{3+} acts as oxidizing agent, the manganese ion attain d^5 electronic configuration

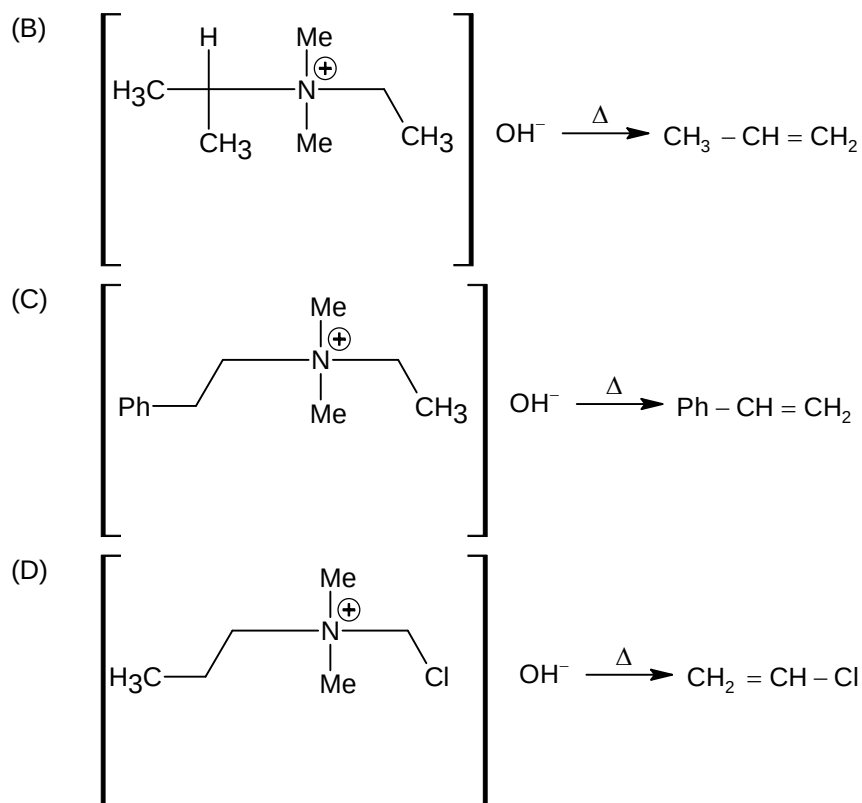
22. For the following sequence, choose the incorrect option(s)?



- (A) IUPAC name of (P) is ethanenitrile
(B) Common name of (S) is acetaldehyde
(C) (Q) is more basic than (P)
(D) (R) can give immediate turbidity with Lucas reagent

23. Which of the following reactions is/are correct regarding the formation of major product (Alkene)?





24. Which of the following is/are related to extraction of metal from silver glance?
- Formation of complex from NaCN solution
 - Air acts as an oxidizing agent
 - Thermal decomposition of cyanide complex
 - Zn-dust acts as reducing agent
25. Two bulbs A and B contains 16 g of O_2 and 16 g O_3 respectively. Which of the following statement(s) is/are true?
- Both bulb contain same number of molecules
 - Both bulb contain same number of atoms
 - Bulb A contains $\frac{N_A}{2}$ molecules while bulb B contains $\frac{N_A}{3}$ molecules. (N_A = Avogadro's number)
 - Both bulb contain 16 g-molecule of respective gases

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

The composition of vapour over a binary ideal solution is determined by the composition of the liquid X_A and Y_A are the mole fractions of 'A' in the liquid and vapour at equilibrium, respectively.

26. The value of X_A for which $(Y_A - X_A)$ is maximum is

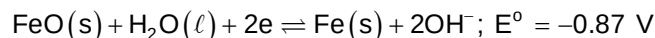
- (A) $\frac{\sqrt{P_A^0 \cdot P_B^0} - P_A^0}{P_A^0 - P_B^0}$
 (B) $\frac{\sqrt{P_A^0 \cdot P_B^0} - P_B^0}{P_A^0 - P_B^0}$
 (C) $\frac{\sqrt{P_A^0 \cdot P_B^0} - P_B^0}{P_A^0}$
 (D) $\frac{\sqrt{P_A^0 \cdot P_B^0} - P_A^0}{P_B^0}$

27. The vapour pressure of solution at this composition

- (A) $\sqrt{P_A^0 \cdot P_B^0}$
 (B) $(P_A^0 - P_B^0)$
 (C) $(P_A^0 + P_B^0)$
 (D) $0.5(P_A^0 + P_B^0)$

Paragraph for Question Nos. 28 and 29

The Edison storage cells represented as $\text{Fe(s)} | \text{FeO(s)} | \text{KOH(aq)} | \text{Ni}_2\text{O}_3(\text{s}) | \text{NiO(s)} | \text{Ni(s)}$. The half-cell reactions are



28. How does cell EMF depend on increasing the concentration of KOH?

- (A) increases
 (B) decreases
 (C) remains unaffected
 (D) none of these

29. What is the maximum amount of electrical energy that can be obtained from one mole of Ni_2O_3 ?

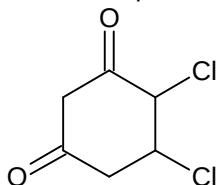
- (A) 2.54 J
 (B) 245.11 kJ
 (C) 122.56 kJ
 (D) 61.28 kJ

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

30. The amount of water produced (in g) in the oxidation of 1 mole of white phosphorus (P_4) with conc. HNO_3 to a compound with highest oxidation state of phosphorus is:

31. Mole fraction of solute in some solution is $1/n$. If 50% of solute molecules dissociate into two parts and remaining 50% get dimerised, new mole fraction of solvent becomes $4/5$. Find value of n .
32. When the following compound is named correctly according to IUPAC convention, what would be the sum of position of two chlorine atoms?



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

KMnO_4 oxidizes X^{n+} ion to XO_3^- in acid solution. 2.5×10^{-3} mole of X^{n+} requires 1.5×10^{-3} mole of MnO_4^-

33. The value of n is _____
34. How many mole of KMnO_4 is needed per mole of X^{n+} to oxidize it to XO_3^- in strong basic medium?

Question Stem for Question Nos. 35 and 36

Question Stem

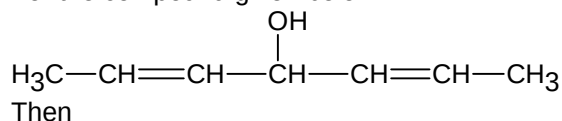
The order of the reaction $\text{A} \longrightarrow \text{Product}$ is zero. The time for completion of reaction is 50 sec. The time for 50% completion of reaction is x sec. and time for 75% completion of reaction is y sec.

35. The value of x is _____
36. The value of y is _____

Question Stem for Question Nos. 37 and 38

Question Stem

If 'x' = Number of diastereomeric pair and
'y' = Number of optical active isomers
For the compound given below



37. The value of x is _____
38. The value of y is _____

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, $S_r = {}^nC_0 \cdot a_r - {}^nC_1 a_{r-1} + {}^nC_2 a_{r-2} - \dots + (-1)^r {}^nC_r \cdot a_0$. Then which of the following statement is/are correct?
 (A) $S_{2020} = 0$
 (B) $S_{2021} = -{}^nC_{673}$
 (C) $S_{2022} = {}^nC_{674}$
 (D) $S_{2019} = -{}^nC_{673}$
40. Four points z_1, z_2, z_3, z_4 in complex plane such that $|z_1| > \frac{4}{3}$, $|z_2| = 1$, $|z_3| \leq 1$ and $z_3 = \frac{z_2(z_1 - z_4)}{\bar{z}_1 z_4 - 1}$, then $|z_4|$ can be equal to
 (A) π
 (B) e
 (C) $\frac{1}{\pi}$
 (D) $\frac{1}{e}$
41. Let a, b, c , be length of side opposite to vertex A, B and C respectively. If $\frac{(s-a)^2}{36} + \frac{(s-b)^2}{9} + \frac{(s-c)^2}{4} = \frac{s^2}{49}$ and $a, b, c \in \mathbb{N}$, HCF of (a, b, c) is 1, where s is the semi-perimeter of the Δ , then which of the following is/are correct?
 (A) Area of ΔABC is 252 sq. unit
 (B) Circumradius of ΔABC is $\frac{55}{42}$
 (C) Semi-perimeter(s) is 49
 (D) The value of $\tan \frac{B}{2}$ is $\frac{4}{7}$
42. Let f be a real valued function such that for any real x . (i) $f(10 + x) = f(10 - x)$, (ii) $f(20 + x) = -f(20 - x)$, then
 (A) f is odd function
 (B) f is even function
 (C) f is periodic function with period 20
 (D) f is periodic function with period 40
43. Let A and B be square matrices of same order which commute with each other such that B is orthogonal and $A^{-1} = A$, $B^{-1} = B$. Suppose $X = \text{adj}(AB^T)^n$, then which of the following is/are correct?
 (Where $\det(AB) > 0$)
 (A) $X = I$ if n is even (when I is identity matrix)
 (B) $X = AB$ if n is even
 (C) $X = BA$ if n is odd
 (D) $X = A^2 B^2$ if n is even

44. Consider the curves S_1 and S_2 where $S_1 : \sin^{-1} x = \cos^{-1} y$ and $S_2 : \tan^{-1} x = \cot^{-1} y$, then
- (A) S_1 and S_2 intersect exactly at 2 points
 - (B) area bounded by S_1 with the coordinate axis is $\frac{\pi}{4}$
 - (C) no of tangents that can be drawn from $(-1, 2)$ to S_2 is 1
 - (D) equation of tangent drawn from point $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ to S_1 is $x + \sqrt{3}y = 2$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

In a plane rectangular co-ordinate system xOy . Points A and B are on the parabola $y^2 = 4x$, satisfying $\overrightarrow{OA} \cdot \overrightarrow{OB} = -4$, where O is the origin and point F is the focus of the parabola

45. Then the product of the area of $\triangle OFA$ and area of the $\triangle OFB$ is
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
46. If range of $FA + FB$ is $[a, \infty)$ and range of $FA \cdot FB$ is $[b, \infty)$, then the value of $a + b$ is
- (A) 15
 - (B) 12
 - (C) 18
 - (D) 24

Paragraph for Question Nos. 47 and 48

If the tangent drawn at a point $(t^2, 2t)$; $t \neq 0$ on the parabola $y^2 = 4x$ is the normal drawn at the point $(\sqrt{5} \cos \phi, 2 \sin \phi)$ on the ellipse $4x^2 + 5y^2 = 20$, then

47. The number of values of ϕ which can be possible ($0 \leq \phi < 2\pi$)
- (A) 3
 - (B) 2
 - (C) 4
 - (D) 1
48. Point of intersection of normal an axis of parabola drawn at point $(t^2, 2t)$ is
- (A) $\left(\frac{11}{5}, 0\right)$
 - (B) $\left(\frac{9}{5}, 0\right)$
 - (C) $\left(\frac{8}{5}, 0\right)$
 - (D) $\left(\frac{12}{5}, 0\right)$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. A person starts from the origin $O(0, 0)$ in the x - y plane. He takes steps of one unit along the positive x -axis or the positive y -axis. Travelling in this manner, the total number of ways he can reach $A(9, 6)$ avoiding both the points $P(3, 3)$ and $Q(6, 4)$
50. The sum of all the solutions of the equation $\cot \frac{\pi x}{2} = \log_2 \{x\}$ in $x \in (0, 100)$ (where $\{.\}$ denotes the fractional part of function) is
51. Let $f(x) = x^3 - 3k^2x + 16$. If $f(|x|) = 0$ has exactly two distinct real roots, then the area bounded by $y = f(x)$ and x -axis in (square unit) is

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53**Question Stem**

If $\int_1^{\infty} \left(\frac{ax}{1+x^2} - \frac{1}{2x} \right) dx = -\frac{\log 2}{k}$ where k is a finite constant

52. The value of a is
53. The value of k is

Question Stem for Question Nos. 54 and 55**Question Stem**

Suppose the real numbers x, y satisfy $x - 4\sqrt{y} = 2\sqrt{x-y}$. Then the range of $x \in [a, b] \cup \{0\}$

54. The value of a is
55. The value of b is

Question Stem for Question Nos. 56 and 57**Question Stem**

Let a, b and c be positive integers consider the class of quadratic equations $ax^2 - bx + c = 0$ having two distinct roots in the open interval $(0, 1)$

56. Then the least integral value of a is
57. Then the least integral value of $a + b + c$ is